

PORTO

RUBA ALMADI

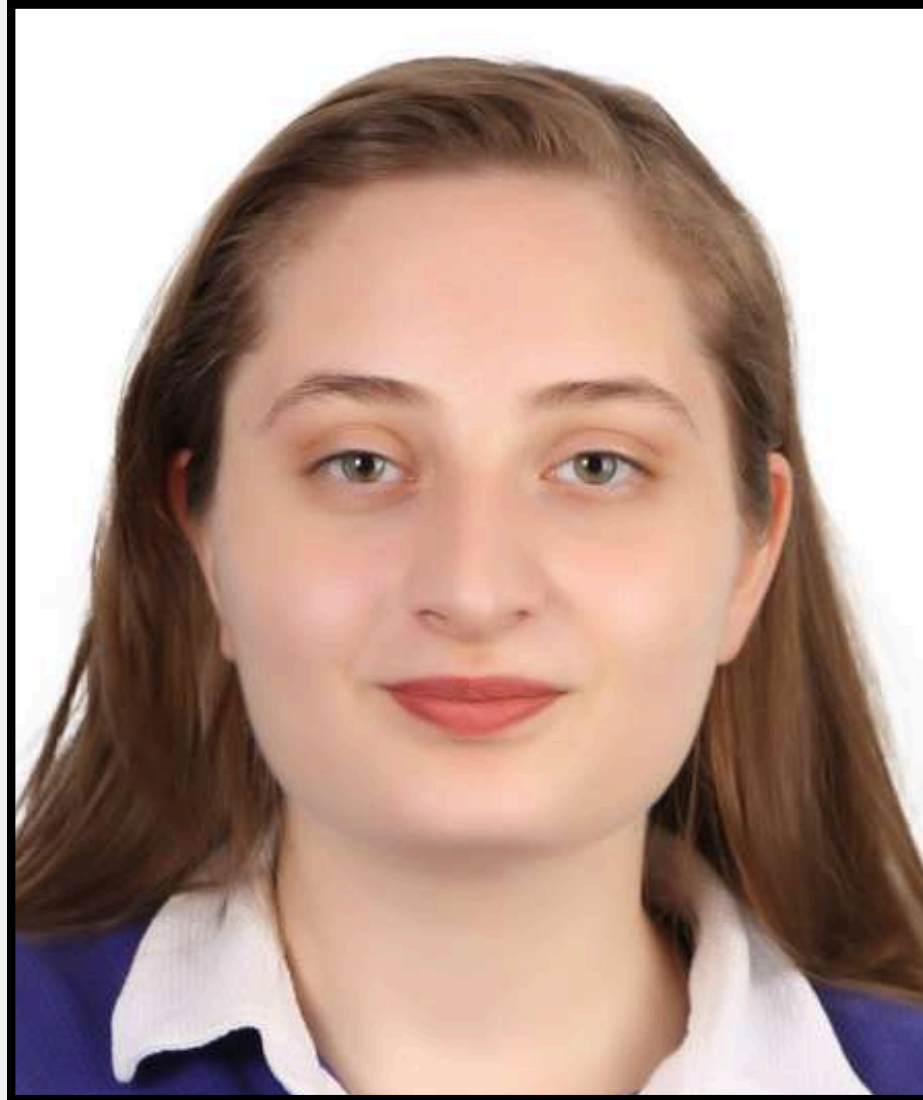
CONTACT INFORMATION

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FOLLO



ABOUT ME

I am a third-year Control and Automation Engineering student at Istanbul Technical University, with a strong focus on dynamic systems, control design, and electromechanical modeling. My work centers on the development of complex engineering systems where control theory, simulation, and physical modeling intersect. In parallel, I lead the Levitation & Control subteam of HyperBee, where I design and validate dynamic models and closed-loop control architectures for electromagnetic levitation systems, integrating control theory with MATLAB/Simulink and FEM-based electromagnetic simulations.

Technical Skills

- **Ansys Maxwell Electronics**
- **MATLAB**
- **SIMULINK**
- **C Programming**
- **Python**
- **LTSpice**

HYPERBEE PROJECT TEAM

HyperBee is a competitive Hyperloop team developing electromagnetic levitation and propulsion systems. I initially worked as a Levitation & Propulsion Team Member, focusing on magnetic circuit design and FEM simulations using ANSYS Maxwell Electronics to optimize levitation performance and vehicle-track interaction. I am currently the Levitation & Control Team Leader, responsible for developing dynamic and electromagnetic models using MATLAB/Simulink and Simscape, as well as designing and validating PID-based closed-loop control architectures to ensure stable and reliable levitation.



COMPETITIONS & ACHIEVEMENTS



- European Hyperloop Week (EHW) is an international Hyperloop competition that brings together university teams from around the world to evaluate levitation, propulsion, and control technologies. As a team, we became the first Turkish Hyperloop team to achieve first place in three different categories, demonstrating strong system-level performance.
- At Teknofest, our team achieved 2nd place in 2024 and received the Steel Levitation Award as the first team to successfully develop and demonstrate a steel-based electromagnetic levitation system. In 2025, we were further recognized with the Electromagnetic System Innovation Award for our electromagnetic levitation system.

AWARDS

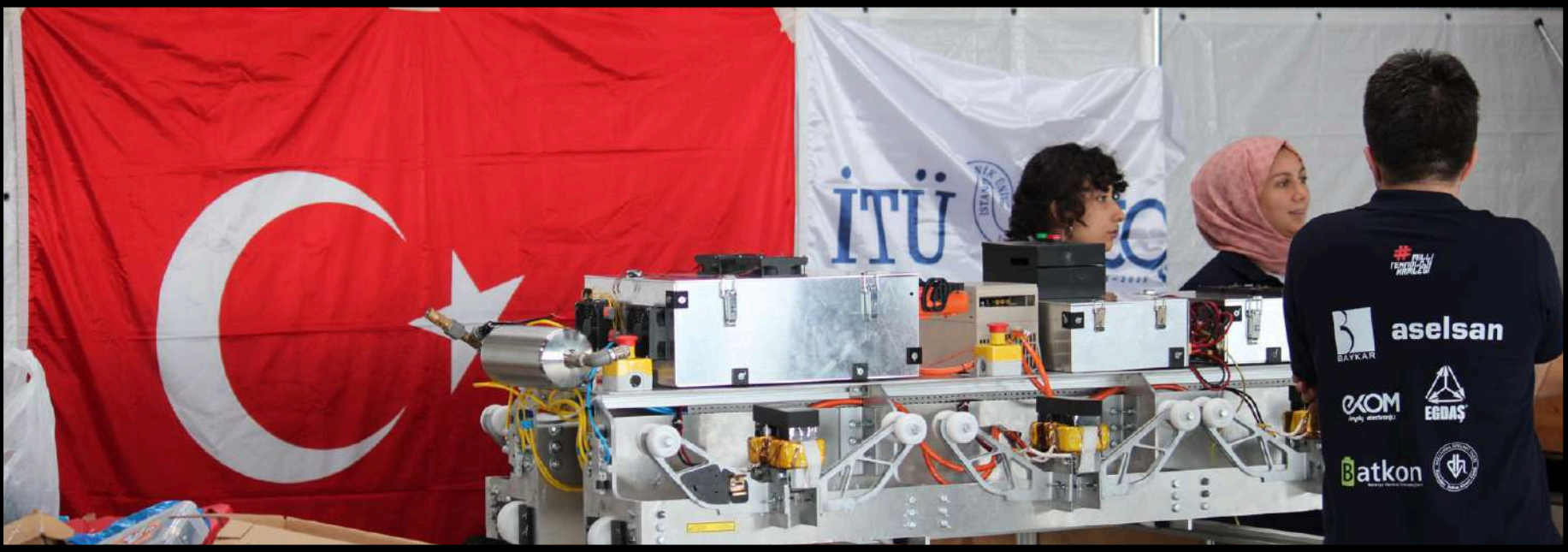
European Hyperloop Week

- Levitation Subsystem Award (2024)
- Propulsion Subsystem Award (2024)
- Sense & Control Subsystem Award (2024)

Teknofest

- 2nd Place Award (2024)
- Steel Levitation Award(2024)
- Electromagnetic System Innovation Award(2025)



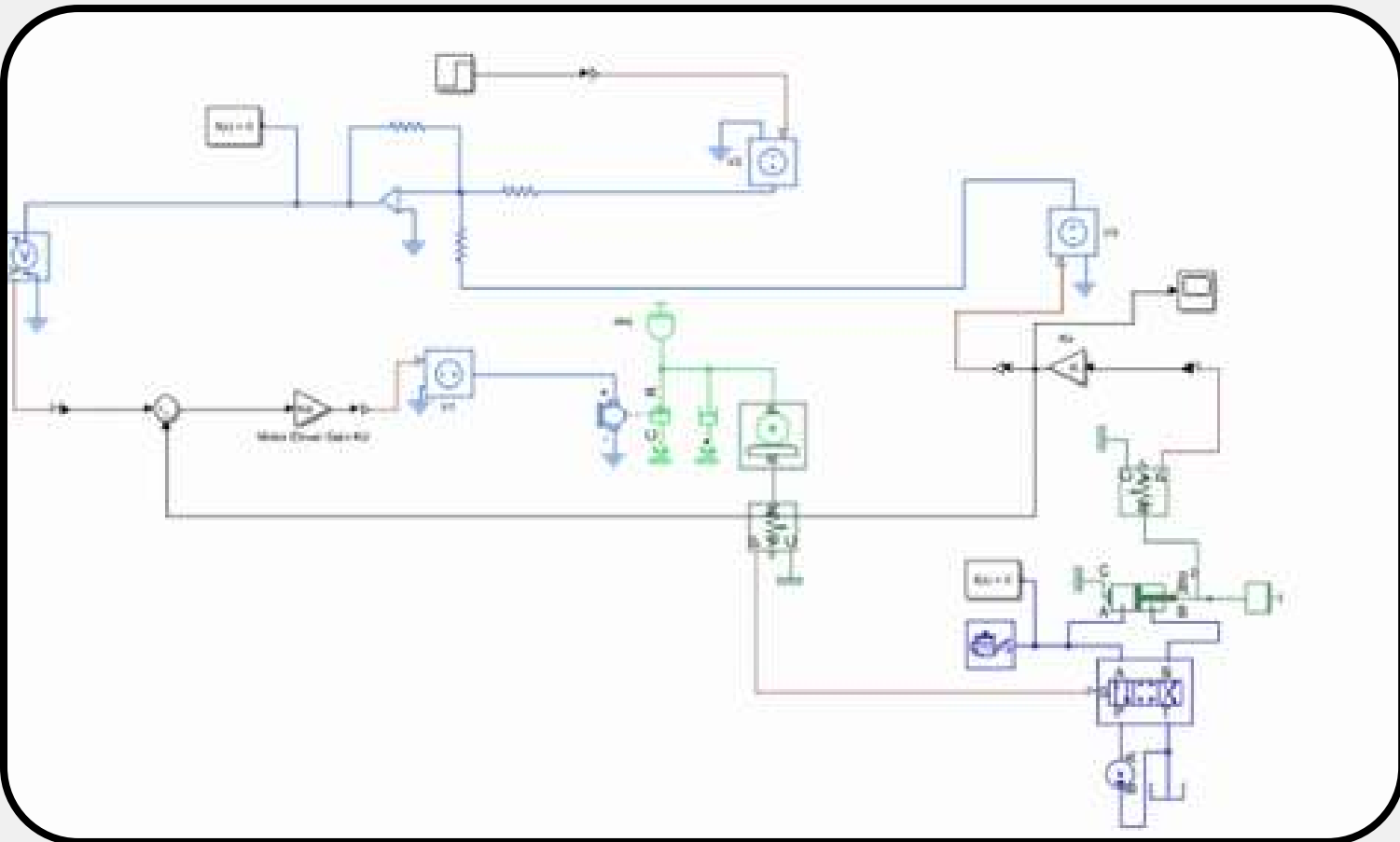


HYPERBEE PROJECT TEAM

OTHER PROJECTS

HYDRAULIC CONVEYOR COMPRESSION SYSTEM

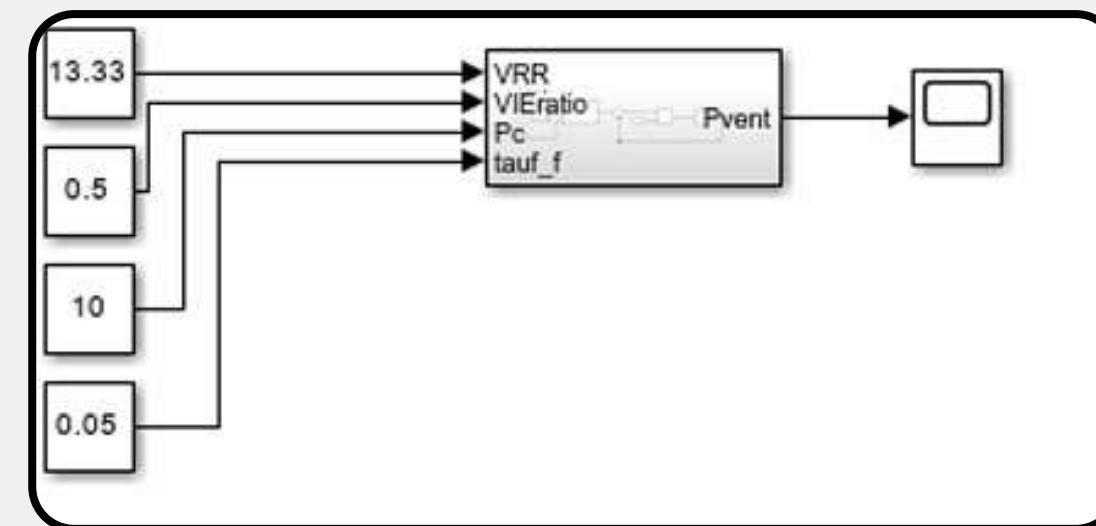
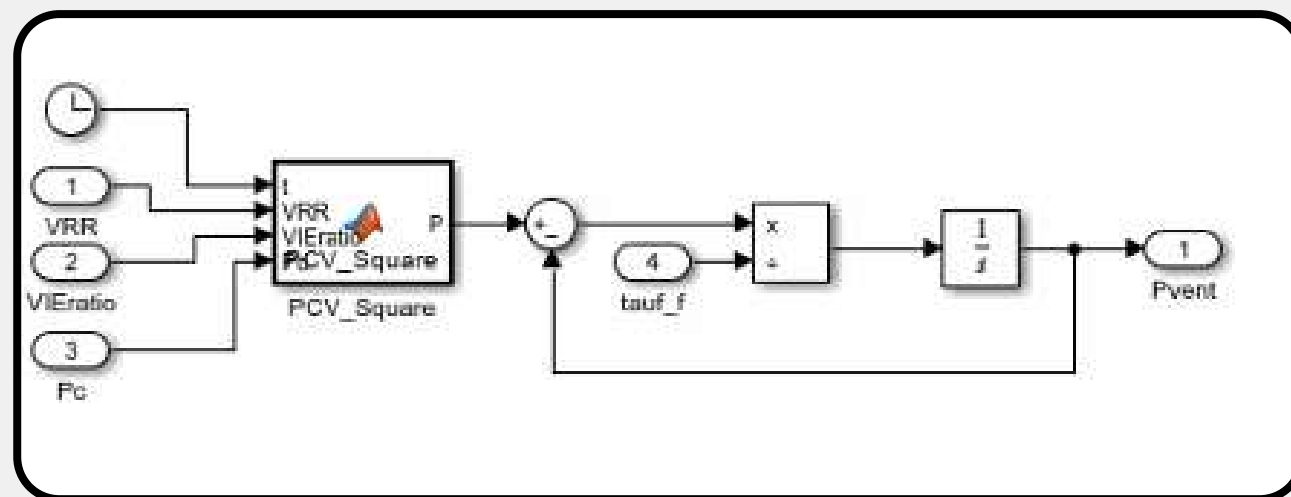
Modeling and simulation of an electromechanical-hydraulic system for bottle compression using a DC motor, valve, and hydraulic piston. The system was modeled using block diagrams and implemented in MATLAB/Simulink and Simscape to analyze dynamic behavior and compare simulation results across modeling approaches.



OTHER PROJECTS

MECHANICAL VENTILATOR CONTROL SYSTEM

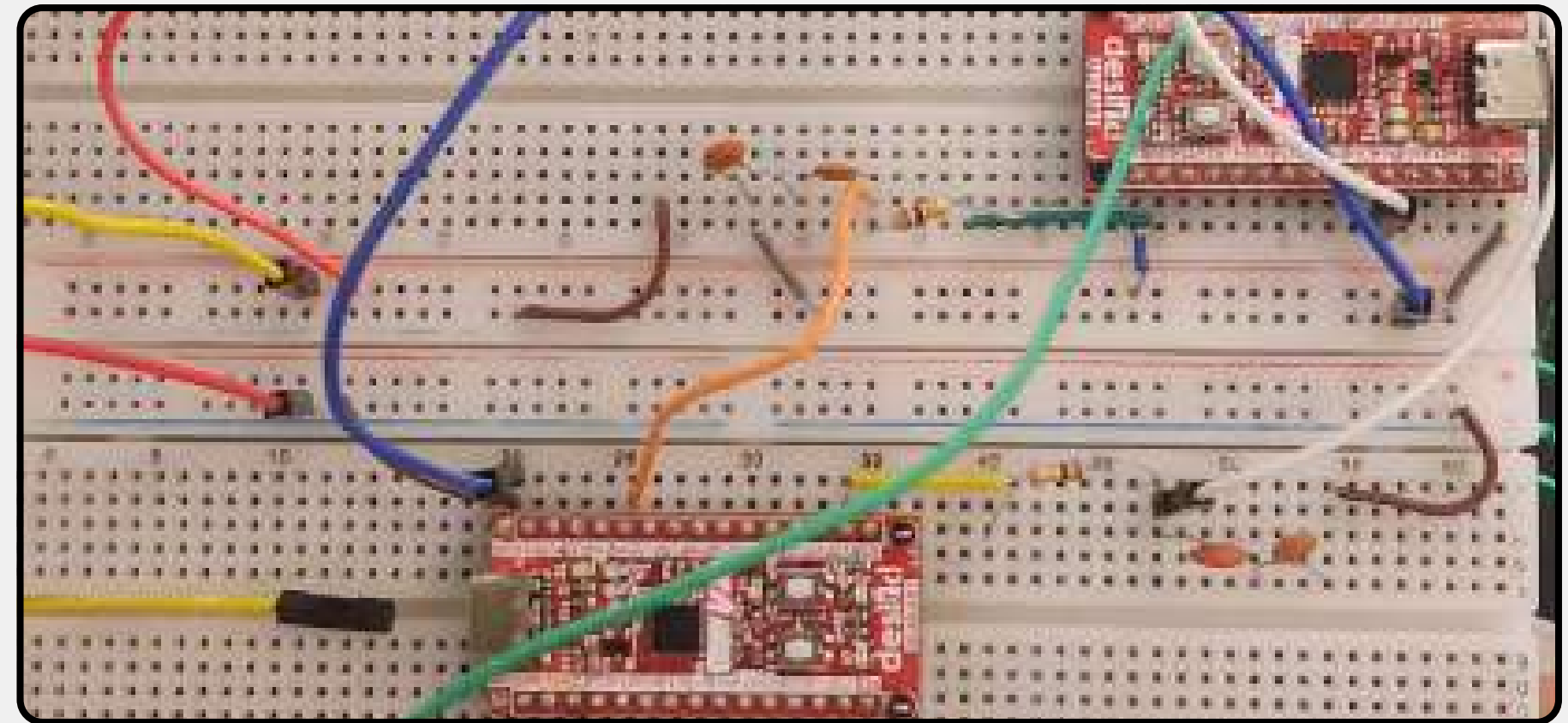
- Design and simulation of a mechanical ventilator system as a control problem. System dynamics and control logic were implemented in MATLAB/Simulink, focusing on control-oriented modeling and algorithmic implementation within a medical device context.



OTHER PROJECTS

Microcontroller Systems – HIL Control Experiment

As part of a hands-on microcontroller systems course, I designed and implemented a control system using a Hardware-in-the-Loop (HIL) approach. The experiment involved developing embedded software on an LPC824 microcontroller to control a continuous-time dynamic plant executed on a separate computer. Control inputs and outputs were exchanged as analog voltages, and the system was discretized at 1 kHz. This work combined low-level embedded programming, peripheral interfacing, and control-oriented implementation, focusing on real-time execution, stability, and closed-loop system behavior.



Duty: 50%	ADC:3098		r: 0. 0		y: 1. 26		e: -1. 26		u: -2. 0	
Duty: 0%	ADC:1964		r: 0. 0		y: -0. 81		e: 0. 81		u: -0.379	
Duty: 40%	ADC:2091		r: 1. 0		y: 0. 42		e: 0.957		u: 2. 0	
Duty:100%	ADC:3062		r: 1. 0		y: 0.990		e: 0. 9		u: 0. 44	
Duty: 51%	ADC:3106		r: 0. 0		y: 1. 33		e: -1. 33		u: -2. 0	
Duty: 0%	ADC:2012		r: 0. 0		y: -0. 34		e: 0. 34		u: 0. 0	
Duty: 50%	ADC:2026		r: 1. 0		y: -0. 21		e: 1. 21		u: 2. 0	
Duty:100%										

Baud

115200

Data rate

8

Parity

none

Handshake

OFF

Mode

Free

THANK

you